

Homework 2 Solutions: Limits and Open Sets

Instructor: Ani Ghosh

Practice Problems: Work together in class

1. For each sequence below, decide whether it converges. If it converges, find its limit.

(a)

$$x_n = \frac{1}{n}.$$

As $n \rightarrow \infty$, the denominator grows without bound, so

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0.$$

Therefore, the sequence converges to 0.

(b)

$$x_n = 5 + \frac{2}{n}.$$

Since

$$\frac{2}{n} \rightarrow 0,$$

we have

$$\lim_{n \rightarrow \infty} \left(5 + \frac{2}{n} \right) = 5.$$

Therefore, the sequence converges to 5.

(c)

$$x_n = n.$$

This sequence grows without bound. Therefore, it does not converge to a finite real number.

(d)

$$x_n = (-1)^n.$$

This sequence alternates between -1 and 1 . Since it does not approach a single number, it does not converge.

2. Suppose the price of a good in period n is given by

$$p_n = 20 + \frac{10}{n}.$$

(a) We compute:

$$p_1 = 20 + \frac{10}{1} = 30,$$

$$p_2 = 20 + \frac{10}{2} = 25,$$

and

$$p_{10} = 20 + \frac{10}{10} = 21.$$

(b) Yes, the price sequence converges.

(c) Since

$$\frac{10}{n} \rightarrow 0,$$

we have

$$\lim_{n \rightarrow \infty} p_n = \lim_{n \rightarrow \infty} \left(20 + \frac{10}{n} \right) = 20.$$

(d) Economically, the price starts above 20 and gradually approaches 20 over time. The value 20 can be interpreted as the long-run price.

3. Suppose a firm's output in period n is

$$q_n = 100 - \frac{20}{n}.$$

(a) We compute:

$$q_1 = 100 - \frac{20}{1} = 80,$$

$$q_2 = 100 - \frac{20}{2} = 90,$$

and

$$q_{20} = 100 - \frac{20}{20} = 99.$$

(b) Yes, the sequence converges.

(c) Since

$$\frac{20}{n} \rightarrow 0,$$

we have

$$\lim_{n \rightarrow \infty} q_n = \lim_{n \rightarrow \infty} \left(100 - \frac{20}{n} \right) = 100.$$

(d) The sequence is increasing. As n rises, $\frac{20}{n}$ becomes smaller, so $100 - \frac{20}{n}$ becomes larger.

4. Consider two consumption bundles

$$\mathbf{x} = (2, 3), \quad \mathbf{y} = (5, 7).$$

Using the Euclidean metric,

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}.$$

Substituting,

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{(2 - 5)^2 + (3 - 7)^2}.$$

Therefore,

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{(-3)^2 + (-4)^2} = \sqrt{9 + 16} = \sqrt{25} = 5.$$

5. Let

$$B_2((1, 1)) = \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x} - (1, 1)\| < 2\}.$$

A point belongs to this ball if its distance from $(1, 1)$ is strictly less than 2.

(a) For $(1, 1)$,

$$\|(1, 1) - (1, 1)\| = 0 < 2.$$

Therefore,

$$(1, 1) \in B_2((1, 1)).$$

(b) For $(2, 1)$,

$$\|(2, 1) - (1, 1)\| = \sqrt{(2-1)^2 + (1-1)^2} = \sqrt{1} = 1 < 2.$$

Therefore,

$$(2, 1) \in B_2((1, 1)).$$

(c) For $(3, 1)$,

$$\|(3, 1) - (1, 1)\| = \sqrt{(3-1)^2 + (1-1)^2} = \sqrt{4} = 2.$$

Since the ball requires distance strictly less than 2, we have

$$(3, 1) \notin B_2((1, 1)).$$

(d) For $(2, 2)$,

$$\|(2, 2) - (1, 1)\| = \sqrt{(2-1)^2 + (2-1)^2} = \sqrt{1+1} = \sqrt{2} < 2.$$

Therefore,

$$(2, 2) \in B_2((1, 1)).$$

Homework Problems: Submit individually

6. For each sequence below, decide whether it converges. If it converges, find its limit.

(a)

$$x_n = \frac{3n+1}{n}.$$

Rewrite:

$$x_n = 3 + \frac{1}{n}.$$

Since $\frac{1}{n} \rightarrow 0$,

$$\lim_{n \rightarrow \infty} x_n = 3.$$

Therefore, the sequence converges to 3.

(b)

$$x_n = \frac{2}{n^2}.$$

Since $n^2 \rightarrow \infty$,

$$\frac{2}{n^2} \rightarrow 0.$$

Therefore, the sequence converges to 0.

(c)

$$x_n = 4 - \frac{1}{n}.$$

Since $\frac{1}{n} \rightarrow 0$,

$$\lim_{n \rightarrow \infty} \left(4 - \frac{1}{n}\right) = 4.$$

Therefore, the sequence converges to 4.

(d)

$$x_n = 2 + (-1)^n.$$

Since $(-1)^n$ alternates between -1 and 1 , the sequence alternates between

$$2 - 1 = 1$$

and

$$2 + 1 = 3.$$

Therefore, the sequence does not converge.

7. Suppose a consumer's planned savings in period n are given by

$$s_n = 500 + \frac{100}{n}.$$

(a) We compute:

$$s_1 = 500 + \frac{100}{1} = 600,$$

$$s_5 = 500 + \frac{100}{5} = 520,$$

and

$$s_{100} = 500 + \frac{100}{100} = 501.$$

(b) Yes, the sequence converges.

(c) Since

$$\frac{100}{n} \rightarrow 0,$$

we have

$$\lim_{n \rightarrow \infty} s_n = \lim_{n \rightarrow \infty} \left(500 + \frac{100}{n} \right) = 500.$$

(d) The limiting value 500 can be interpreted as the consumer's long-run planned savings level. Planned savings start above 500 and gradually approach 500 over time.

8. Let

$$S = (0, 10) = \{x \in \mathbb{R} : 0 < x < 10\}.$$

Interpret S as the set of feasible prices for a good.

(a) Yes, S is open in $\mathbb{R}$. For every point $x \in (0, 10)$, we can find a small open interval around x that remains inside $(0, 10)$.

(b) No, S is not closed. It does not contain its boundary points 0 and 10.

(c) The boundary points of S are

$$0 \quad \text{and} \quad 10.$$

(d) No,

$$0 \notin S.$$

(e) No,

$$10 \notin S.$$

9. Let

$$T = [0, 10] = \{x \in \mathbb{R} : 0 \leq x \leq 10\}.$$

(a) No, T is not open in $\mathbb{R}$. For example, any open interval around 0 contains negative numbers, which are not in T .

(b) Yes, T is closed in $\mathbb{R}$. It contains its boundary points 0 and 10.

(c) The boundary points of T are

$$0 \quad \text{and} \quad 10.$$

(d) One example of a sequence contained in T that converges to 10 is

$$x_n = 10 - \frac{1}{n}.$$

For every $n \geq 1$,

$$x_n \in [0, 10],$$

and

$$\lim_{n \rightarrow \infty} x_n = 10.$$

(e) Yes, the limit belongs to T , since

$$10 \in [0, 10].$$

10. Let the feasible set of consumption bundles be

$$X = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 > 0, x_2 > 0\}.$$

(a) Yes,

$$(1, 1) \in X,$$

because both coordinates are strictly positive.

(b) No,

$$(0, 1) \notin X,$$

because the first coordinate is not strictly positive.

(c) Yes, X is open in $\mathbb{R}^2$. Every point in X has both coordinates strictly positive, so we can draw a sufficiently small open ball around that point that remains inside X .

(d) No, X is not closed in $\mathbb{R}^2$. It does not contain boundary points such as $(0, 1)$.

(e) The sequence is

$$\mathbf{x}_n = \left(\frac{1}{n}, 1 \right).$$

For every positive integer n ,

$$\frac{1}{n} > 0 \quad \text{and} \quad 1 > 0.$$

Therefore,

$$\mathbf{x}_n \in X$$

for every n .

(f) The limit is

$$\lim_{n \rightarrow \infty} \mathbf{x}_n = \lim_{n \rightarrow \infty} \left(\frac{1}{n}, 1 \right) = (0, 1).$$

(g) The limit does not belong to X , because the first coordinate is 0, not strictly positive. This shows that X is not closed: it contains a convergent sequence whose limit is not contained in the set.